	Ciron Drugs & Pharmaceuticals Pvt. Ltd. N-118,119,113,119/1,119/2 & 118/1, M.I.D.C., Tarapur, Boisar, Dist. Palghar - 401 506.

Item Description	**Leaflet**	**Artwork Number**	**N/A**
Product Name	**HYDRALAT (Hydralazine Hydrochloride Injection)**	**Customer / Country**	**Zulat Pharmacy**
Artwork Code	**N/A**	**Pack Size**	**5 x 1ml Ampoule**
Substrate Specification	**60 GSM Maplitho Paper**	**Pharmacode / Barcode**	**N/A**
Minimum Font Size	**5.5 pt**	**Dimension**	**215 x 154mm**
Colour Scheme (Pantone Number)	**Black**	**Artist Name**	**Pritesh**
		Language	**English**
Subject	**N/A**	**Date**	**10-04-2025**
Reason For Revision	**N/A**	**Printer Name**	**N/A**

215 mm

1 Front 2

154 mm

HYDRALAT

Hydralazine Hydrochloride Injection 20mg/ml

Composition:
Each ml contains:
Hydralazine Hydrochloride 20mg

Preservatives:
Methyl Paraben 0,95mg
Propyl Paraben 0,35mg

Dosage Form
Parenteral, Injection, liquid

Product Description
Clear colourless liquid filled in amber coloured ampoules. Amber coloured glass ampoule (1 ml) was packed in a tray as 5 or 10 ampoule in duplex board carton as 10 x 1ml or 5 x 1ml Amp.

Pharmacodynamics
Hydralazine is a peripheral vasodilator.
Mechanism of action:
Hydralazine is a direct acting vasodilator which exerts its effects principally on the arterioles. Its precise mode of action is not known. Administration of hydralazine produces a fall in peripheral resistance and a decrease in arterial blood pressure, effects which induce reflex sympathetic cardiovascular responses. The concomitant use of beta-blocker will reduce these reflex effects and enhance the antihypertensive effect. The use of hydralazine can result in sodium and fluid retention, producing oedema and reduced urinary volume. These effects can be prevented by concomitant administration of a diuretic.

Pharmacokinetics
Absorption: None stated

Distribution:
Hydralazine is rapidly distributed in the body and displays a particular affinity for the blood- vessel walls. Plasma protein binding is of the order of 90%. Biotransformation: None stated

Elimination:
Plasma half-life average 2-3 hours but is prolonged up to 16 hours in severe renal failure (creatinine clearance less than 20ml/min) and shortened to approximately 45minutes in rapid acetylators.

Characteristic in patients: None stated
Indication
Treatment of hypertensive emergencies, particularly those associated with preeclampsia and toxaemia of pregnancy,
Treatment of hypertension with renal complications.

Recommended Dose
Adults:
When there is urgent need, therapy in the hospitalized patient may be initiated intramuscularly or as a rapid intravenous bolus injection directly into the vein, Hydralazine hydrochloride injection should be used only when the drug cannot be given orally. The usual dose is 20 to 40 mg, repeated as necessary. Certain patients (especially those with marked renal damage) may require a lower dose. Blood pressure should be checked frequently. It may begin to fall within a few minutes after injection, with the average maximal decrease occurring in 10 to 80 minutes. In cases where there has been increased intracranial pressure, lowering the blood pressure may increase cerebral ischemia. Most patients can be transferred to oral hydralazine hydrochloride within 24 to 48 hours.

The product should be used immediately after the vial is opened. It should not be added to infusion solutions. Hydralazine hydrochloride injection may discolour upon contact with metal; discoloured solutions should be discarded.
Parenteral drug products should be inspected visually for particulate matter and discoloration prior to administration, whenever solution and container permit.

Children:
Not recommended

Elderly:
Clinical evidence would indicate that no special dosage regime is necessary. Advancing age does not affect either blood concentration or systemic clearance.
Renal elimination may however be affected in so far as kidney function diminishes with age.

Route Of Administration
Intramuscular / Intravenous.

Contraindication
Known hypersensitivity to hydralazine or Di hydralazine.
Idiopathic systemic lupus erythematosus (SLE) and related diseases.
Severe tachycardia and heart failure with a high cardiac output (e.g., in thyrotoxicosis). Myocardial insufficiency due to mechanical obstruction (e.g., in the presence of aortic or mitral stenosis or constrictive pericarditis).
Isolated right ventricular failure due to pulmonary hypertension (cor pulmonale). Dissecting aortic aneurysm.

Warning And Precautions
Warning
The overall 'hyperdynamic' state of the circulation induced by hydralazine may accentuate certain clinical conditions. Myocardial stimulation may provoke or aggravate angina pectoris. Patients with suspected or confirmed coronary artery disease should therefore be given Hydralazine HCl Injection only under cover beta-blocker or in combination with other suitable sympatholytic agents. It is important that the beta-blocker medication should be commenced a few days before the start of treatment with Hydralazine HCl Injection

Patients who have survived a myocardial infarction should not receive Hydralazine HC1 Injection until a post- infarction stabilization phase has been achieved.
Prolonged treatment with hydralazine (i.e., usually for more than 6 months) may provoke a lupus erythematosus (LE) like syndrome, especially where doses exceed 100 mg daily. First symptoms are likely to be arthralgia, sometimes associated with fever and rash and are reversible after withdrawal of the drug.
In its more severe form, it resembles acute SLE, and in rare cases renal and ocular involvements have been reported. Long-term treatment with corticosteroids may be required to reverse these changes. Since such reactions tend to occur more frequently the higher the dose and the longer its duration, and since they are more common in slow acetylators, it is recommended that for maintenance therapy the lowest effective dose should be used. If 100 mg daily fails to elicit an adequate clinical effect, the patient's acetylator status should be evaluated. Slow acetylators and women run greater risk of developing the LE like syndrome and every effort should therefore be made to keep the dosage below 100 mg daily and a careful watch kept for signs and symptoms suggestive of this syndrome. If such symptoms do develop the drug should be gradually withdrawn. Rapid acetylators often respond inadequately even to doses of 100 mg daily and therefore the dose can be raised with only a slightly increased risk of an LE-like syndrome.
During long term treatment with Hydralazine HCl Injection it is advisable to determine the antinuclear factors and conduct urine analysis at intervals of approximately 6 months. Microhematuria and / or proteinuria, in particular together with positive titres of ANF, may be initial signs of immune-complex glomerulonephritis associated with the SLE like syndrome. If overt clinical signs or symptoms develop, the drug should be withdrawn immediately. Skin rash, febrile reactions and change in blood count occur rarely and drug should be withdrawn. Peripheral neuritis in the form of paresthesia has been reported and may respond to pyridoxine administration or drug withdrawal. In high (cyto-) toxic concentrations, hydralazine induces gene mutations in single cell organisms and in mammalian cells in vitro. No unequivocally mutagenic effects have been detected in vivo in a great number of test systems. Hydralazine in lifetime carcinogenicity studies, caused, towards the end of the experiments, small but statistically significant increases in lung tumors in mice and in hepatic and testicular tumors in rats. These tumors also occur spontaneously with fairly high frequency in aged rodents. With due consideration of these animals and in-vitro toxicological findings, hydralazine in therapeutic doses does not appear to bear risk that would necessitate a limitation of its administration. Many years of clinical experience have not suggested that human cancer is associated with hydralazine use.

Precautions
In patients with renal impairment (creatine clearance < 30 ml/min or serum creatinine concentrations> 2.5 mg / 100 ml or 221 Mmol / 1) and in patients with hepatic dysfunction the dose or interval between doses should be adjusted according to clinical response, in order to avoid accumulation of the 'apparent' active substance.

Hydralazine HC1 Injection should be used with caution in patients with coronary artery disease (since it may increase angina) or cerebrovascular disease. When undergoing surgery, patients treated with Hydralazine HC1 Inj may show a fall in blood pressure, in which case one should not use adrenaline to correct the hypotension, since it enhances the cardiac-accelerating effects of hydralazine.

Interactions With Other Medicaments
Potentiation of effects: Concurrent therapy with other antihypertensives (vasodilators, calcium antagonists, ACE inhibitors, diuretics), anaesthetics tricyclic antidepressants, major tranquilizers or drugs exerting central depressant actions (including alcohol).
Administration of Hydralazine HC1 Inj shortly before or after diazoxide may give rise to marked hypotension.
MAO inhibitors should be used with caution in patients receiving Hydralazine HCl Inj. Concurrent administration of Hydralazine HC1 Inj with beta-blockers subject to a strong first pass effect (e.g., propranolol) may increase their bioavailability. Download adjustment of these drugs may be required when they are given concomitantly with Hydralazine HC1 Inj.

Pregnancy And Lactation
Use of Hydralazine HC1 Inj in pregnancy, before the third trimester should be avoided but the drug maybe employed in later pregnancy if there is no safer alternative or when the disease itself carries serious risks for the mother or child e.g., pre-eclampsia and /or eclampsia.
No serious adverse effects in human pregnancy have been reported to date with Hydralazine HCl Inj, although experience in the third trimester is extensive.

Breast-Feeding:
Hydralazine passes into breast milk but reports available so far have not shown adverse effects on the infant. Mothers in whom use of Hydralazine HCl Inj proves unavoidable may breast feed their infant provided that the infant is observed for possible adverse effects.

Side Effects
Some of the adverse effects listed below e.g. tachycardia, palpitation, angina symptoms, flushing, headache, dizziness, nasal congestion and gastro-intestinal disturbances are commonly seen at the start of treatment, especially if the dose is raised quickly.

However, such effects generally subside in the further course of treatment. (The following frequency estimates are used: frequent> 10 %, occasional 1-10% rare 0.001 - 1% isolated cases < 0.001%)

Cardiovascular system:
Frequently: tachycardia, palpitation.
Occasionally: flushing, hypotension, anginal symptoms.
Rarely: oedema, heart failure.
Isolated cases: paradoxical pressor responses.

Central and peripheral nervous system:
Frequently: headache.
Rarely: dizziness.
Isolated cases: peripheral neuritis, polyneuritis, paraesthesiae (these unwanted effects may be reversed by administering pyridoxine).

Musculo-skeletal system:
Occasionally: arthralgia, joint swelling, myalgia.

Skin and appendages:
Rarely: rash.

Urogenital system:
Rarely: proteinuria, increased plasma creatinine, haematuria sometimes in association with glomerulonephritis.
Isolated cases: acute renal failure, urinary retention.

Gastro-intestinal tract:
Occasionally: gastro-intestinal disturbances, diarrhoea, nausea, vomiting.
Rarely: jaundice, liver enlargement, abnormal liver function sometimes in association with hepatitis.
Isolated cases: paralytic ileus.

Blood:
Rarely: anaemia, leucopenia, neutropenia, thrombocytopenia with or without purpura.
Isolated cases: haemolytic anaemia, leucocytosis, lymphadenopathy, pancytopenia, splenomegaly, agranulocytosis.

Psyche:
Rarely: agitation, anorexia, anxiety.
Isolated cases: depression, hallucinations.

Sense organs:
Rarely: increased lacrimination, conjunctivitis, nasal congestion.

Hypersensitivity reactions:
Occasionally: SLE-like syndrome (see under 'Warnings')
Rarely: hypersensitivity reactions such as pruritus, urticaria, vasculitis, eosinophilia, hepatitis.

Respiratory tract:
Rarely: dyspnoea, pleural pain.

Miscellaneous:
Rarely: fever, weight decrease, malaise.
Isolated cases: exophthalmos

Symptoms And Treatment of Overdose
Signs and symptoms
Symptoms include hypotension, tachycardia, myocardial ischaemia dysrhythmias and coma.

Treatment
Supportive measures including intravenous fluids are also indicated. If hypotension is present, an attempt should be made to raise the blood pressure without increasing the tachycardia. Adrenaline should therefore be avoided.

Effects on ability to drive and use machines:
Hydralazine may impair the patient's reactions especially at the start of the treatment. The patient should be warned of the hazard when driving or operating machinery

Storage Condition
Store in a dry place, below 30°C and protect from light.

Shelf Life
24 Months from the date of manufacture. Single used: Discard any unused portion.

Presentations
Pack 5 x 1m and 10 x 1m of glass ampoule.

Manufactured By:
Ciron Drugs & Pharmaceuticals Pvt. Ltd.
N-118, 118/1,119,119/1,119/2,113 M.I.D.C.
Tarapur, Boisar, Palghar-401506.
Maharashtra State, India.

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Zulat Pharmacy Sdn. Bhd
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Date Of Revision
7th July 2023.

3 Back 4

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